

ADVANCED ALGORITHMS: PART 1

1. Consider a distributed consensus network operating under the Raft consensus protocol. When a network partition splits 5 nodes into a cluster of 3 and a cluster of 2, explain how the leader election proceeds and which partition commits new transactions.

Continuation of Question 1 Choices:

- A. The minority cluster of 2 elects a leader and continues writing.
- B. The majority cluster of 3 maintains quorum and commits transactions.
- C. Both partitions halt immediately.
- D. Neither partition can achieve consensus.

2. Which sorting algorithm has an average time complexity of $O(N \log N)$ and is typically implemented in-place?

- A. Merge Sort
- B. Quick Sort
- C. Radix Sort
- D. Counting Sort